

IB MATH AA SL

## Lesson 31

# The Binomial Distribution

A large, stylized, light blue letter 'X' is positioned on the right side of the slide. It is set against a dark blue background that features two large, overlapping circles of a slightly lighter shade of blue. The 'X' is composed of two intersecting diagonal lines with thick strokes.

Statistics & Probability · 60-minute lesson

Syllabus SL 4.8 · Property of Lynda Badmus Education

# Do Now — recall

## ON YOUR WHITEBOARD

1.  $P(\text{heads})$  in one coin toss?
2. In 10 tosses, how many heads do you expect?
3. Is each coin toss independent?

## Answers (click to reveal)

1.  $\frac{1}{2}$
2. 5
3. yes

# What we're learning today

## LEARNING OBJECTIVES

- 1** **Recognise**  
a binomial situation.
- 2** **Use**  
 $X \sim B(n, p)$ .
- 3** **Find**  
binomial probabilities with the GDC.
- 4** **Find**  
the mean  $E(X) = np$ .

## SUCCESS CRITERIA — I CAN...

- ✓  
check the binomial conditions
- ✓  
write  $X \sim B(n, p)$
- ✓  
find  $P(X = r)$  and  $P(X \leq r)$  on the GDC
- ✓  
use mean  $np$  and variance  $np(1-p)$

# When is it binomial?

$X \sim B(n, p)$

$$X \sim B(n, p)$$

**n**

fixed number of trials

**p**

same probability of success each trial

**Trials**

independent, two outcomes (success/fail)

## FORMULAS

**Mean**

$$E(X) = np$$

**Variance**

$$\text{Var}(X) = np(1-p)$$

**GDC**

BinomPdf / BinomCdf.

# A binomial probability

## QUESTION

A fair coin is tossed 6 times. Find  
P(exactly 4 heads).  
Here  $X \sim B(6, 0.5)$ .

## SOLUTION

Use the binomial formula (or GDC BinomPdf):

$$P(X=4) = \binom{6}{4} (0.5)^4 (0.5)^2$$

$$= 15 \times (0.5)^6:$$

$$= \frac{15}{64} \approx 0.234$$

# Your turn — binomial

## ANSWER IN YOUR BOOK

1.  $X \sim B(10, 0.5)$ : expected value?
2.  $X \sim B(20, 0.3)$ : mean  $np$ ?
3. State the two parameters of  $B(n, p)$ .

## Answers (click to reveal)

1. 5
2. 6
3.  $n$  (trials),  $p$  (success prob.)

# Mean & cumulative

## USING THE GDC

$$E(X) = np$$

**P(X = r)**

BinomPdf(n, p, r)

**P(X ≤ r)**

BinomCdf(n, p, r)

**P(X ≥ r)**

1 - P(X ≤ r-1)

## MEAN & VAR

**Mean**

np

**Variance**

np(1 - p)

**SD**

$\sqrt{np(1-p)}$ .

# Mean of a binomial

## QUESTION

For  $X \sim B(50, 0.2)$ ,  
find the expected value  $E(X)$ .

## SOLUTION

Binomial mean  $E(X) = np$ :

$$E(X) = 50 \times 0.2$$

Evaluate:

$$E(X) = 10$$

# Exam-style question

In a quiz, the probability of guessing a question correctly is 0.25. A student guesses all 12 questions.  $X \sim B(12, 0.25)$ .

- (a) Find the expected number correct. [2]
- (b) Find  $P(X = 3)$ . [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a)  $E(X) = np = 12 \times 0.25 = 3$  (M1)(A1)
- (b)  $\text{BinomPdf}(12, 0.25, 3) \approx 0.258$  (M1)(A1)

# Common mistakes



## Conditions

Binomial needs fixed  $n$ , constant  $p$ , independent trials.



## Pdf vs Cdf

$P(X = r)$  uses Pdf;  $P(X \leq r)$  uses Cdf.



## 'At least'

$P(X \geq r) = 1 - P(X \leq r - 1)$ .



## Mean

$E(X) = np$ , not  $n + p$ .

# Mixed practice

## QUESTIONS

1.  $X \sim B(8, 0.5)$ : mean?
2.  $X \sim B(100, 0.1)$ : mean?
3.  $P(X \geq 1)$  in terms of  $P(X=0)$ ?
4. Variance of  $B(20, 0.5)$ ?

## Answers (click to reveal)

1. 4
2. 10
3.  $1 - P(X=0)$
4.  $20 \cdot 0.5 \cdot 0.5 = 5$

# Plenary & exit ticket

## KEY FORMULAS

$$X \sim B(n, p)$$

$$E(X) = np, \quad \text{Var} = np(1-p)$$

## EXIT TICKET — before you leave

1. Mean of  $B(40, 0.25)$ ?
2.  $P(X \leq r)$ : Pdf or Cdf?
3. Two parameters of binomial?

### Answers (click to reveal)

1. 10
2. Cdf
3. n and p

# Homework

## SET ON DR FROST MATHS (auto-marked)

### Task:

Binomial distribution.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

## NEXT LESSON

### Lesson 32

The normal distribution — the bell curve.

$$X \sim N(\mu, \sigma^2)$$